DO NOT WRITE IN THIS AREA
DO NOT WRITE IN

15 The straight line L_1 has equation $y = 3x - 4$
The straight line L_2 is perpendicular to L_1 and passes through the point (9, 5)

Find an equation of line L_2

L_1 has gradient 3; perpendicular gradient = −1/3.

L_2: $y = -x/3 + c$; $5 = -9/3 + c$, so $c = 8$.

Hence $y = -x/3 + 8$.

3 marks total
Perpendicular −1/3
Substitute (9,5)
y = −x/3 + 8

(Total for Question 15 is 3 marks)